COMPUTER ENGINEERING

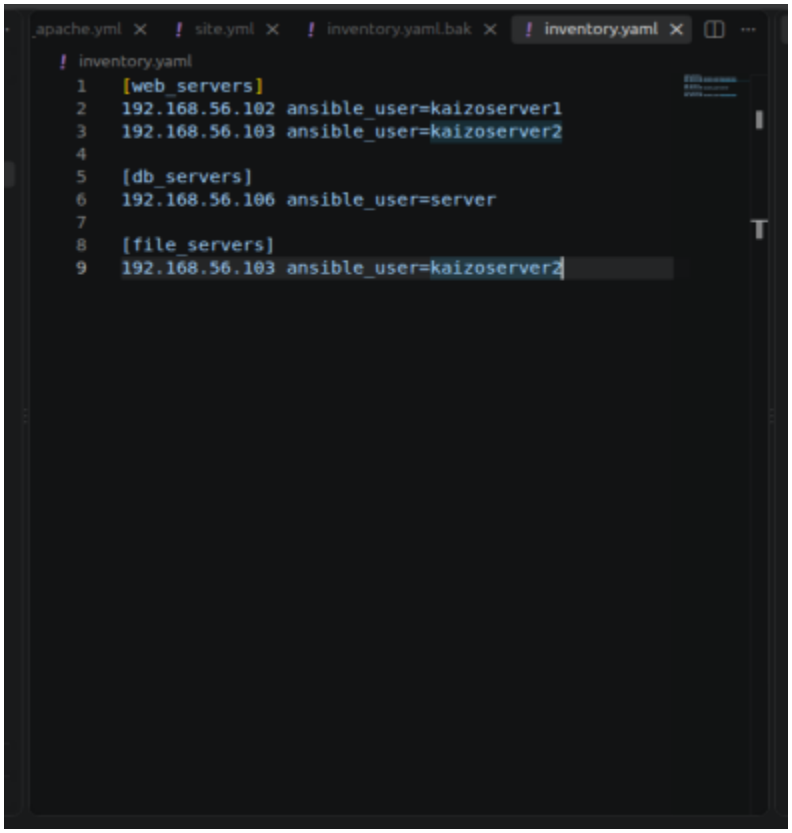
QUEZON CITY

COURSE CODE/TITLE:	CPE 212	**SECTION:**	CPE31S2
DATE PERFORMED:	09/06/26	**DATE SUBMITTED:**	09/06/26
NAME:	Ivan Wayne San Juan	**INSTRUCTOR:**	Engr. Jimlord Quejado

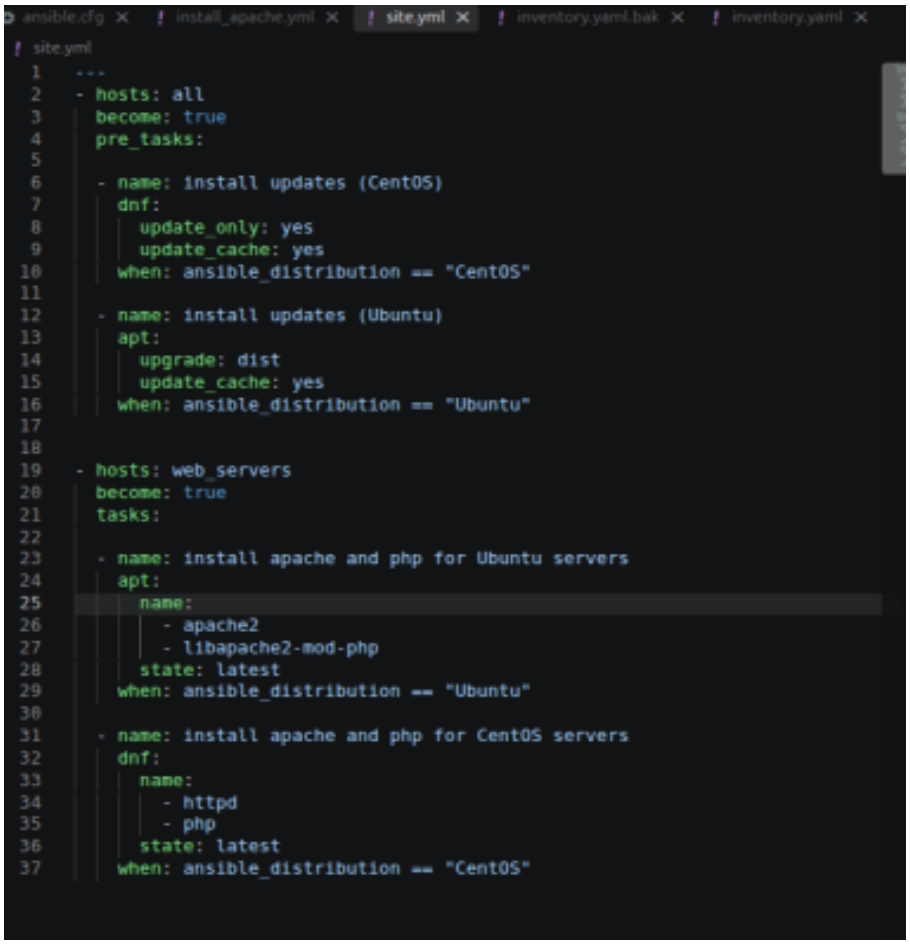
ACTIVITY NO. 6.1 Targeting Specific Nodes.

1. PROCEDURE OUTPUT

1.2



1.3


ansible.cfg
install_apache.yml
site.yml
inventory.yaml.bak
inventory.yaml
site.yml

- hosts: all
become: true
pre_tasks:
- name: install updates (CentOS)
dnf:
update_only: yes
update_cache: yes
when: ansible_distribution == "CentOS"
- name: install updates (Ubuntu)
apt:
upgrade: dist
update_cache: yes
when: ansible_distribution == "Ubuntu"
- hosts: web_servers
become: true
tasks:
- name: install apache and php for Ubuntu servers
apt:
name:
- apache2
- libapache2-mod-php
state: latest
when: ansible_distribution == "Ubuntu"
- name: install apache and php for CentOS servers
dnf:
name:
- httpd
- php
state: latest
when: ansible_distribution == "CentOS"

```
katzohost@katzohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ ansible-playbook -b -K site.yml
BECOME password:

PLAY [all] *********************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [192.168.56.102]
ok: [192.168.56.103]
ok: [192.168.56.106]

TASK [install updates (CentOS)] ************************************************************************************
skipping: [192.168.56.102]
skipping: [192.168.56.103]
ok: [192.168.56.106]

TASK [install updates (Ubuntu)] ************************************************************************************
skipping: [192.168.56.106]
changed: [192.168.56.102]
changed: [192.168.56.103]

PLAY [web_servers] *************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [192.168.56.103]
ok: [192.168.56.102]

TASK [install apache and php for Ubuntu servers] *******************************************************************
ok: [192.168.56.102]
ok: [192.168.56.103]

TASK [install apache and php for CentOS servers] *******************************************************************
skipping: [192.168.56.102]
skipping: [192.168.56.103]

PLAY RECAP *********************************************************************************************************
192.168.56.102             : ok=4    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.103             : ok=4    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.106             : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
```

1.4

```
- hosts: db_servers
  become: true
  tasks:

  - name: install mariadb package (CentOS)
    yum:
      name: mariadb-server
      state: latest
    when: ansible_distribution == "CentOS"

  - name: "Mariadb- Restarting/Enabling"
    service:
      name: mariadb
      state: restarted
      enabled: true

  - name: install mariadb package (Ubuntu)
    apt:
      name: mariadb-server
      state: latest
    when: ansible_distribution == "Ubuntu"
```

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ ansible-playbook -b -K site.yml
BECOME password:

PLAY [all] *********************************************************************

TASK [Gathering Facts] *********************************************************
ok: [192.168.56.103]
ok: [192.168.56.102]
ok: [192.168.56.106]

TASK [install updates (CentOS)] ************************************************
skipping: [192.168.56.102]
skipping: [192.168.56.103]
ok: [192.168.56.106]

TASK [install updates (Ubuntu)] ************************************************
skipping: [192.168.56.106]
ok: [192.168.56.103]
ok: [192.168.56.102]

PLAY [web_servers] *************************************************************

TASK [Gathering Facts] *********************************************************
ok: [192.168.56.102]
ok: [192.168.56.103]

TASK [install apache and php for Ubuntu servers] *******************************
ok: [192.168.56.103]
ok: [192.168.56.102]

TASK [install apache and php for CentOS servers] *******************************
skipping: [192.168.56.102]
skipping: [192.168.56.103]

PLAY [db_servers] **************************************************************
```

```
PLAY [db_servers] **************************************************************

TASK [Gathering Facts] *********************************************************
ok: [192.168.56.106]

TASK [install mariadb package (CentOS)] ****************************************
changed: [192.168.56.106]

TASK [Mariadb- Restarting/Enabling] ********************************************
changed: [192.168.56.106]

TASK [install mariadb package (Ubuntu)] ****************************************
skipping: [192.168.56.106]

PLAY RECAP *********************************************************************
192.168.56.102             : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.103             : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.106             : ok=5    changed=2    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
```

1.5
CentOS

```
● mariadb.service - MariaDB 10.5 database server
     Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: disabled)
     Active: active (running) since Sat 2026-09-05 22:20:45 PST; 6min ago
       Docs: man:mariadbd(8)
             https://mariadb.com/kb/en/library/systemd/
    Process: 7171 ExecStartPre=/usr/libexec/mariadb-check-socket (code=exited, status=0/SUCCESS)
    Process: 7193 ExecStartPre=/usr/libexec/mariadb-prepare-db-dir mariadb.service (code=exited, status=0>
    Process: 7301 ExecStartPost=/usr/libexec/mariadb-check-upgrade (code=exited, status=0/SUCCESS)
   Main PID: 7281 (mariadbd)
     Status: "Taking your SQL requests now..."
      Tasks: 8 (limit: 22979)
     Memory: 81.4M (peak: 100.7M)
        CPU: 449ms
     CGroup: /system.slice/mariadb.service
             └─7281 /usr/libexec/mariadbd --basedir=/usr

Sep 05 22:20:45 localhost.localdomain mariadb-prepare-db-dir[7233]: The second is mysql@localhost, it has>
Sep 05 22:20:45 localhost.localdomain mariadb-prepare-db-dir[7233]: you need to be the system 'mysql' use>
Sep 05 22:20:45 localhost.localdomain mariadb-prepare-db-dir[7233]: After connecting you can set the pass>
Sep 05 22:20:45 localhost.localdomain mariadb-prepare-db-dir[7233]: able to connect as any of these users>
Sep 05 22:20:45 localhost.localdomain mariadb-prepare-db-dir[7233]: See the MariaDB Knowledgebase at http>
Sep 05 22:20:45 localhost.localdomain mariadb-prepare-db-dir[7233]: Please report any problems at https:/>
Sep 05 22:20:45 localhost.localdomain mariadb-prepare-db-dir[7233]: The latest information about MariaDB >
Sep 05 22:20:45 localhost.localdomain mariadb-prepare-db-dir[7233]: Consider joining MariaDB's strong and>
Sep 05 22:20:45 localhost.localdomain mariadb-prepare-db-dir[7233]: https://mariadb.org/get-involved/
Sep 05 22:20:45 localhost.localdomain systemd[1]: Started MariaDB 10.5 database server.
~
```

Ubuntu

```
kaizoserver1@kaizoserver1-VirtualBox:~$ systemctl status mariadb
Unit mariadb.service could not be found.
```

The mariadb in CentOS is enabled but running and in ubuntu it's literally off

1.6

```
66
67  - hosts: file_servers
68    become: true
69    tasks:
70      - name: install samba package
71        package:
72          name: samba
73          state: latest
```

2.1

site.yml

```yaml
---
- hosts: all
  become: true
  pre_tasks:

  - name: install updates (CentOS)
    tags: always
    dnf:
      update_only: yes
      update_cache: yes
    when: ansible_distribution == "CentOS"

  - name: install updates (Ubuntu)
    tags: always
    apt:
      upgrade: dist
      update_cache: yes
    when: ansible_distribution == "Ubuntu"


- hosts: web_servers
  become: true
  tasks:

  - name: install apache and php for Ubuntu servers
    tags: apache, apache2, ubuntu
    apt:
      name:
        - apache2
        - libapache2-mod-php
      state: latest
    when: ansible_distribution == "Ubuntu"

  - name: install apache and php for CentOS servers
    tags: apache, centos, httpd
    dnf:
      name:
        - httpd
        - php
      state: latest
```

```yaml

  - name: install apache and php for Ubuntu servers
    tags: apache, apache2, ubuntuS
    apt:
      name:
        - apache2
        - libapache2-mod-php
      state: latest
    when: ansible_distribution == "Ubuntu"

  - name: install apache and php for CentOS servers
    tags: apache, centos, httpd
    dnf:
      name:
        - httpd
        - php
      state: latest
    when: ansible_distribution == "CentOS"

- hosts: db_servers
  become: true
  tasks:

  - name: install mariadb package (CentOS)
    tags: centos, db,mariadb
    yum:
      name: mariadb-server
      state: latest
    when: ansible_distribution == "CentOS"

  - name: "Mariadb- Restarting/Enabling"
    service:
      name: mariadb
      state: restarted
      enabled: true

  - name: install mariadb package (Ubuntu)
```

```yaml
- hosts: web_servers
  - name: install apache and php for CentOS servers
    dnf:
      name:
        - php
      state: latest
    when: ansible_distribution == "CentOS"

- hosts: db_servers
  become: true
  tasks:

  - name: install mariadb package (CentOS)
    tags: centos, db,mariadb
    yum:
      name: mariadb-server
      state: latest
    when: ansible_distribution == "CentOS"

  - name: "Mariadb- Restarting/Enabling"
    service:
      name: mariadb
      state: restarted
      enabled: true

  - name: install mariadb package (Ubuntu)
    tags: db, mariadb,ubuntu
    apt:
      name: mariadb-server
      state: latest
    when: ansible_distribution == "Ubuntu"

- hosts: file_servers
  become: true
  tasks:
    - name: install samba package
      tags: samba
      package:
        name: samba
        state: latest
```

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ ansible-playbook -b -K site.yml
BECOME password:

PLAY [all] *********************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [192.168.56.102]
ok: [192.168.56.103]
ok: [192.168.56.106]

TASK [install updates (CentOS)] ************************************************************************************
skipping: [192.168.56.102]
skipping: [192.168.56.103]
ok: [192.168.56.106]

TASK [install updates (Ubuntu)] ************************************************************************************
skipping: [192.168.56.106]
ok: [192.168.56.102]
ok: [192.168.56.103]

PLAY [web_servers] *************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [192.168.56.103]
ok: [192.168.56.102]

TASK [install apache and php for Ubuntu servers] *******************************************************************
ok: [192.168.56.102]
ok: [192.168.56.103]

TASK [install apache and php for CentOS servers] *******************************************************************
skipping: [192.168.56.102]
skipping: [192.168.56.103]

PLAY [db_servers] **************************************************************************************************
```

```
ok: [192.168.56.103]
ok: [192.168.56.102]

TASK [install apache and php for Ubuntu servers] ********************************************************************
ok: [192.168.56.102]
ok: [192.168.56.103]

TASK [install apache and php for CentOS servers] ********************************************************************
skipping: [192.168.56.102]
skipping: [192.168.56.103]

PLAY [db_servers] ***************************************************************************************************

TASK [Gathering Facts] **********************************************************************************************
ok: [192.168.56.106]

TASK [install mariadb package (CentOS)] *****************************************************************************
ok: [192.168.56.106]

TASK [Mariadb- Restarting/Enabling] *********************************************************************************
changed: [192.168.56.106]

TASK [install mariadb package (Ubuntu)] *****************************************************************************
skipping: [192.168.56.106]

PLAY [file_servers] *************************************************************************************************

TASK [Gathering Facts] **********************************************************************************************
ok: [192.168.56.103]

TASK [install samba package] ****************************************************************************************
ok: [192.168.56.103]

PLAY RECAP **********************************************************************************************************
192.168.56.102             : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.103             : ok=6    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.106             : ok=5    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
```

2.2.1

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ ansible-playbook --list-tags site.yml

playbook: site.yml

  play #1 (all): all    TAGS: []
      TASK TAGS: [always]

  play #2 (web_servers): web_servers    TAGS: []
      TASK TAGS: [apache, apache2, centos, httpd, ubuntu]

  play #3 (db_servers): db_servers      TAGS: []
      TASK TAGS: [centos, db, mariadb, ubuntu]

  play #4 (file_servers): file_servers  TAGS: []
      TASK TAGS: [samba]
```

2.2.2

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ ansible-playbook --tags centos -K -b site.yml
BECOME password:

PLAY [all] *********************************************************************

TASK [Gathering Facts] *********************************************************
ok: [192.168.56.102]
ok: [192.168.56.103]
ok: [192.168.56.106]

TASK [install updates (CentOS)] ************************************************
skipping: [192.168.56.102]
skipping: [192.168.56.103]
ok: [192.168.56.106]

TASK [install updates (Ubuntu)] ************************************************
skipping: [192.168.56.106]
ok: [192.168.56.102]
ok: [192.168.56.103]

PLAY [web_servers] *************************************************************

TASK [Gathering Facts] *********************************************************
ok: [192.168.56.103]
ok: [192.168.56.102]

TASK [install apache and php for CentOS servers] *******************************
skipping: [192.168.56.102]
skipping: [192.168.56.103]

PLAY [db_servers] **************************************************************

TASK [Gathering Facts] *********************************************************
ok: [192.168.56.106]

TASK [install mariadb package (CentOS)] ****************************************
ok: [192.168.56.106]

PLAY [file_servers] ************************************************************

TASK [install apache and php for CentOS servers] *******************************
skipping: [192.168.56.102]
skipping: [192.168.56.103]

PLAY [db_servers] **************************************************************

TASK [Gathering Facts] *********************************************************
ok: [192.168.56.106]

TASK [install mariadb package (CentOS)] ****************************************
ok: [192.168.56.106]

PLAY [file_servers] ************************************************************

TASK [Gathering Facts] *********************************************************
ok: [192.168.56.103]

PLAY RECAP *********************************************************************
192.168.56.102             : ok=3    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.103             : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.106             : ok=4    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
```

2.2.3

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ ansible-playbook --tags db -K -b site.yml
BECOME password:

PLAY [all] *********************************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************
ok: [192.168.56.102]
ok: [192.168.56.103]
ok: [192.168.56.106]

TASK [install updates (CentOS)] ************************************************************************************************************************
skipping: [192.168.56.102]
skipping: [192.168.56.103]
ok: [192.168.56.106]

TASK [install updates (Ubuntu)] ************************************************************************************************************************
skipping: [192.168.56.106]
ok: [192.168.56.103]
ok: [192.168.56.102]

PLAY [web_servers] *************************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************
ok: [192.168.56.102]
ok: [192.168.56.103]

PLAY [db_servers] **************************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************
ok: [192.168.56.106]

TASK [install mariadb package (CentOS)] ****************************************************************************************************************
ok: [192.168.56.106]

TASK [install mariadb package (Ubuntu)] ****************************************************************************************************************
skipping: [192.168.56.106]
```

```
TASK [install mariadb package (CentOS)] ****************************************************************************************************************
ok: [192.168.56.106]

TASK [install mariadb package (Ubuntu)] ****************************************************************************************************************
skipping: [192.168.56.106]

PLAY [file_servers] ************************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************
ok: [192.168.56.103]

PLAY RECAP *********************************************************************************************************************************************
192.168.56.102             : ok=3    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.103             : ok=4    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.106             : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
```

2.2.4

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ ansible-playbook --tags apache -K -b site.yml
BECOME password:

PLAY [all] *********************************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************
ok: [192.168.56.102]
ok: [192.168.56.106]
ok: [192.168.56.103]

TASK [install updates (CentOS)] ************************************************************************************************************************
skipping: [192.168.56.102]
skipping: [192.168.56.103]
ok: [192.168.56.106]

TASK [install updates (Ubuntu)] ************************************************************************************************************************
skipping: [192.168.56.106]
ok: [192.168.56.103]
ok: [192.168.56.102]

PLAY [web_servers] *************************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************
ok: [192.168.56.102]
ok: [192.168.56.103]

TASK [install apache and php for Ubuntu servers] *******************************************************************************************************
ok: [192.168.56.102]
ok: [192.168.56.103]

TASK [install apache and php for CentOS servers] *******************************************************************************************************
skipping: [192.168.56.102]
skipping: [192.168.56.103]

PLAY [db_servers] **************************************************************************************************************************************
```

```
PLAY [db_servers] ***************************************************************************************************************************************

TASK [Gathering Facts] **********************************************************************************************************************************
ok: [192.168.56.106]

PLAY [file_servers] *************************************************************************************************************************************

TASK [Gathering Facts] **********************************************************************************************************************************
ok: [192.168.56.103]

PLAY RECAP **********************************************************************************************************************************************
192.168.56.102             : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.103             : ok=5    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.106             : ok=3    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
```

2.2.5

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ ansible-playbook --tags "apache,db" -b -K site.yml
BECOME password:

PLAY [all] **********************************************************************************************************************************************

TASK [Gathering Facts] **********************************************************************************************************************************
ok: [192.168.56.102]
ok: [192.168.56.103]
ok: [192.168.56.106]

TASK [install updates (CentOS)] *************************************************************************************************************************
skipping: [192.168.56.102]
skipping: [192.168.56.103]
ok: [192.168.56.106]

TASK [install updates (Ubuntu)] *************************************************************************************************************************
skipping: [192.168.56.106]
ok: [192.168.56.102]
ok: [192.168.56.103]

PLAY [web_servers] **************************************************************************************************************************************

TASK [Gathering Facts] **********************************************************************************************************************************
ok: [192.168.56.103]
ok: [192.168.56.102]

TASK [install apache and php for Ubuntu servers] ********************************************************************************************************
ok: [192.168.56.103]
ok: [192.168.56.102]

TASK [install apache and php for CentOS servers] ********************************************************************************************************
skipping: [192.168.56.102]
skipping: [192.168.56.103]

PLAY [db_servers] ***************************************************************************************************************************************

TASK [Gathering Facts] **********************************************************************************************************************************
ok: [192.168.56.106]
```

```
PLAY [web_servers] **************************************************************************************************************************************

TASK [Gathering Facts] **********************************************************************************************************************************
ok: [192.168.56.103]
ok: [192.168.56.102]

TASK [install apache and php for Ubuntu servers] ********************************************************************************************************
ok: [192.168.56.103]
ok: [192.168.56.102]

TASK [install apache and php for CentOS servers] ********************************************************************************************************
skipping: [192.168.56.102]
skipping: [192.168.56.103]

PLAY [db_servers] ***************************************************************************************************************************************

TASK [Gathering Facts] **********************************************************************************************************************************
ok: [192.168.56.106]

TASK [install mariadb package (CentOS)] *****************************************************************************************************************
ok: [192.168.56.106]

TASK [install mariadb package (Ubuntu)] *****************************************************************************************************************
skipping: [192.168.56.106]

PLAY [file_servers] *************************************************************************************************************************************

TASK [Gathering Facts] **********************************************************************************************************************************
ok: [192.168.56.103]

PLAY RECAP **********************************************************************************************************************************************
192.168.56.102             : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.103             : ok=5    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.106             : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
```

3.1

```
42
43    - name: start httpd (CentOS)
44      tags: apache,centos,httpd
45      service:
46        name: httpd
47        state: started
48      when: ansible_distribution == "CentOS"
49
```

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ ansible-playbook -b -K site.yml
BECOME password:

PLAY [all] *********************************************************************

TASK [Gathering Facts] *********************************************************
ok: [192.168.56.103]
ok: [192.168.56.102]
ok: [192.168.56.106]

TASK [install updates (CentOS)] ************************************************
skipping: [192.168.56.102]
skipping: [192.168.56.103]
ok: [192.168.56.106]

TASK [install updates (Ubuntu)] ************************************************
skipping: [192.168.56.106]
ok: [192.168.56.102]
ok: [192.168.56.103]

PLAY [web_servers] *************************************************************

TASK [Gathering Facts] *********************************************************
ok: [192.168.56.102]
ok: [192.168.56.103]

TASK [install apache and php for Ubuntu servers] *******************************
ok: [192.168.56.102]
ok: [192.168.56.103]

TASK [install apache and php for CentOS servers] *******************************
skipping: [192.168.56.102]
skipping: [192.168.56.103]

TASK [start httpd (CentOS)] ****************************************************
```

```
TASK [install apache and php for CentOS servers] *******************************
skipping: [192.168.56.102]
skipping: [192.168.56.103]

TASK [start httpd (CentOS)] ****************************************************
skipping: [192.168.56.102]
skipping: [192.168.56.103]

PLAY [db_servers] **************************************************************

TASK [Gathering Facts] *********************************************************
ok: [192.168.56.106]

TASK [install mariadb package (CentOS)] ****************************************
ok: [192.168.56.106]

TASK [Mariadb- Restarting/Enabling] ********************************************
changed: [192.168.56.106]

TASK [install mariadb package (Ubuntu)] ****************************************
skipping: [192.168.56.106]

PLAY [file_servers] ************************************************************

TASK [Gathering Facts] *********************************************************
ok: [192.168.56.103]

TASK [install samba package] ***************************************************
ok: [192.168.56.103]

PLAY RECAP *********************************************************************
192.168.56.102             : ok=4    changed=0    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
192.168.56.103             : ok=6    changed=0    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
192.168.56.106             : ok=5    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ ansible-galaxy init roles/webserver
- Role roles/webserver was created successfully
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ ansible-galaxy init roles/database
- Role roles/database was created successfully
```

2

roles > webserver > tasks > ! main.yml

```yaml
---
- name: install apache and php for Ubuntu servers
  apt:
    name:
      - apache2
      - libapache2-mod-php
    state: latest
  when: ansible_distribution == "Ubuntu"

- name: install apache and php for CentOS servers
  dnf:
    name:
      - httpd
      - php
    state: latest
  when: ansible_distribution == "CentOS"
```

3

roles > webserver > tasks > ! main.yml

```yaml
---
- name: install apache and php for Ubuntu servers
  apt:
    name:
      - apache2
      - libapache2-mod-php
    state: latest
  when: ansible_distribution == "Ubuntu"

- name: install apache and php for CentOS servers
  dnf:
    name:
      - httpd
      - php
    state: latest
  when: ansible_distribution == "CentOS"
```

4

```
17
18  - name: install mariadb package (CentOS)
19    yum:
20      name: mariadb-server
21      state: latest
22    when: ansible_distribution == "CentOS"
23
24  - name: "Mariadb- Restarting/Enabling"
25    service:
26      name: mariadb
27      state: restarted
28      enabled: true
29
30  - name: install mariadb package (Ubuntu)
31    apt:
32      name: mariadb-server
33      state: latest
34    when: ansible_distribution == "Ubuntu"
```

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ ansible-playbook -K -b roles_sites.yml
BECOME password:

PLAY [web_servers] ***********************************************************************************************************************************

TASK [Gathering Facts] *******************************************************************************************************************************
ok: [192.168.56.102]
ok: [192.168.56.103]

TASK [webserver : install apache and php for Ubuntu servers] *****************************************************************************************
ok: [192.168.56.102]
ok: [192.168.56.103]

TASK [webserver : install apache and php for CentOS servers] *****************************************************************************************
skipping: [192.168.56.102]
skipping: [192.168.56.103]

PLAY [db_servers] ************************************************************************************************************************************

TASK [Gathering Facts] *******************************************************************************************************************************
ok: [192.168.56.106]

PLAY RECAP *******************************************************************************************************************************************
192.168.56.102             : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.103             : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.106             : ok=1    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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